Younsoo Park

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Education

The Pennsylvania State University University Park, PA

Bachelor of Science in Computer Science Minor in Mathematics Expected Graduation: May 2027 Dean's List

Research Interests

Systems Security
Network Security
Distributed Systems
LLM Systems
AI Agent Memory
Federated Learning
TinyML for IoT Security
Adversarial Machine Learning
Edge AI Systems
Autonomous Driving
Uncertainty-Aware Motion Prediction

Research Experience

Undergraduate Researcher Oct 2026 – Present

Diffusion-Residual Tube MPC for Autonomous Driving (WIP, targeting ITEC)

Research focus: Uncertainty-aware motion prediction that gives a Tube MPC planner scene-dependent safety regions around surrounding vehicles.

Key contributions:

- Built an Argoverse 2 Motion Forecasting pipeline over 224,896 scenarios (765,659 vehicle tracks) with constant-velocity, constant-acceleration, and CTRV baselines.
- Trained a GRU nominal predictor with nearby lane centerlines and a residual diffusion model; switching from epsilon to v-prediction fixed a sampler that amplified errors about 2000×.
- Calibrated 90% occupancy ellipses with split conformal prediction; at the same coverage the 3 s regions average 22.4 m^2, 34% smaller than a fixed Gaussian (33.9 m^2) and 64% smaller than a conformal circle (62.7 m^2).
- Ran the predictor with a Tube MPC (CasADi / IPOPT) in CARLA 0.9.16 at 10 Hz, recalibrated on CARLA traffic.
- Over 160 closed-loop runs (4 cut-in scenarios × 2 models × 20 seeds), the ego drove 44% faster than a constant-velocity, fixed-Gaussian baseline (36.9 vs 25.7 km/h) through 271 cut-ins.

Undergraduate Research Assistant & Team Lead Oct 2025 – Present

Reliable Federated TinyML Deployment for IoT Security (WIP, LCTES '26)

Penn State University and Elizabethtown College Supervisors: Dr. Suman Saha and Dr. Peilong Li

Research focus: Federated and adversarially robust TinyML systems for IoT security and resource-constrained edge-device deployment.

Key contributions:

- Designed a federated learning pipeline using FedAvgM, focal loss, and cosine learning-rate scheduling on CIC-IDS2017.
- Improved attack recall from 46.7% to 93.85% and F1-score from 84.1% to 89.32%.
- Built a TinyML compression pipeline including BatchNorm folding, structured pruning, knowledge distillation, and INT8 QAT/PTQ.
- Achieved 12.28× model compression and 74.5% inference latency reduction for ESP32 deployment.
- Evaluated adversarial robustness using PGD attacks (ε-sweep 0.01–0.2) across 48 compression configurations.

Publications and Manuscripts

Diffusion-Residual Tube MPC for Autonomous Driving Work in Progress Target Venue: ITEC

Reliable Federated TinyML Deployment for IoT Security Work in Progress Target Venue: ACM LCTES 2026

Independent Study

AI Agent Memory Systems Jul 2026 – Present

Advisor: Prof. Kiwan Maeng

- Researched five production and research AI agent memory systems (Mem0, A-MEM, LangMem, Zep, OpenAI Memory) to identify the SOTA approach, comparing memory creation, retrieval, and lifecycle-management approaches.
- Identified a core architectural divergence: Mem0 extracts and retrieves discrete facts, while A-MEM models memory as an evolving, interconnected knowledge structure; traced how Mem0's open-source implementation diverged from its published ADD/UPDATE/DELETE design into an ADD-only, retrieval-time resolution approach.
- Narrowed research focus to Mem0 (implementation depth, practical extensibility) and A-MEM (architectural novelty) for application to the lab's production AI agent system.
- Applying memory architectures to production recommendation systems through a Memory-first Personalization framework, shipped as Wowness at Levit Inc.

Industry and Technical Experience

Levit Inc. (Shopport AI) Seoul, South Korea

Associate Problem Solver June 2026 – Aug 2026

- Worked directly with the CEO to define product direction, prioritize engineering initiatives, and translate customer problems into production AI systems for an LLM-powered shopping assistant.
- Built a production evaluation platform analyzing **49,807 conversations** and **80,124 query-level routing decisions**, achieving **97.7% evaluation coverage** through versioned routing audit and historical backfill pipelines that reprocessed 47,231 conversations spanning 100+ days of production history.
- Developed end-to-end experimentation infrastructure across frontend, backend, workers, and admin systems, improving recommendation completion from **62.8%** to **75.0% (+12.2 percentage points)**.

- Evolved Shopport's routing stack through three architectural generations – LLM classification, an embedding-based semantic fast path ($\sim 8\times$ faster for confident cases), and a hybrid classifier for ambiguous queries – reaching **86% V3 routing accuracy** and rescuing **93%** of eligible router errors.
- Designed and shipped Wowness, a Mem0-backed cross-category discovery recommender grounded in the live catalog, as sole engineer across a 919-line NestJS engine, React UI, and admin review lab – extending the Memory-first Personalization architecture from proposal to production.
- Diagnosed production failures at the system boundary, including a split MongoDB connection that held the V3 feature switch at zero executions, and corrected systematic analytics false positives, reducing one queue-level count from 2,005 to 811.
- Contributed **730 GitHub commits** and **143 merged pull requests**, adding over **114,000 lines** of TypeScript/TSX during the engagement.

SIGNUM (Nittany AI Alliance – PIT-UN Funded) Aug 2024 – Nov 2025

Co-founder of a $3,750 PIT-UN funded AI platform for caregiver decision support.

- Built multi-source ETL pipelines integrating CMS, NPPES, and Google Places datasets into a DuckDB warehouse.
- Developed hospital-rating prediction models using Markov Transition analysis with confidence intervals.
- Designed and deployed domain-specific RAG systems using FAISS and AWS Bedrock, reducing caregiver research time from 50+ hours per week to near-instant answers.

Atom Tech Solutions LTD Berkeley, CA

Software Engineering Intern May 2022 – Aug 2022

- Built secure authentication systems using SQL/JWT authentication and encrypted credential storage.
- Implemented automated credential generation and authenticated onboarding workflows.

Technical Projects

GladIEEEators – Penn State IEEE Battle Bots (1st Place) Apr 2026

- Designed Shot & Chaser, a coordinated two-robot combat system, for the IEEE Student Chapter Battle Bots Competition, winning 1st place.
- Engineered a belt-driven AR500 vertical-spinner weapon system (334 J at 18,500 RPM, 200+ mph tip speed), validated via Fusion FEA above 4,000 N with a 2.09 minimum safety factor.
- Designed mixed-material armor (6061 aluminum, UHMW, TPU) and integrated drivetrain, radio control, and protected LiPo electronics across both robots within a $976 total build budget.

XIIO – LinkedIn for Filmmakers Apr 2026 – Present

- Built a creator networking platform featuring portfolios, video streaming, collaboration workflows, and school-based discovery using Next.js, Firebase, and Cloudflare Stream.
- Designed real-time messaging, creator profiles, media uploads, and production portfolio management.

ASME @ Penn State Website Aug 2024 – Present

- Developed the official ASME Penn State website using React, TypeScript, Vite, and Firebase.
- Designed a multi-tier permission and approval system for organization management.

Hangukgwan, Taiwan, Software Engineer Summer 2022, Summer 2025

- Built and run the live ordering, POS, and back-office system for a two-store business (650+ commits, 211 test files, about 42K lines); every push to main deploys to the open stores.
- Shipped QR table ordering that prints kitchen tickets unattended: replaced 2–4 s polling with Pusher push events and moved printing into a Java Android POS app that sends ESC/POS jobs to the kitchen printer over TCP after Android Chrome blocked browser printing.
- Built payroll from photographed punch cards, read in the browser with digit templates (no staff data leaves the device), with hourly and monthly pay and overtime proposed for the owner to confirm.
- Seeded an ingredient ledger with 20,877 rows (NT$14.76M) of the owner's spreadsheets for 21 months of price trends; supplier receipts are read with the Claude API (about $2.50/month) after on-device OCR stalled at 22% line accuracy.
- Delivered monthly and annual P&L (revenue minus ingredients and labor) from daily close records, pushed to the owner on LINE, plus ECPay checkout, split tender, refunds, VIP discounts, and a four-language Next.js site.

Teaching Experience

Lead Teaching Assistant, Math 230 (Multivariable Calculus) Penn State University Aug 2025 – Present

Responsibilities include assisting instruction, supporting students during office hours, and helping coordinate course activities.

Leadership and Service

THON Technology Committee Technology Captain (2027) Apr 2026 – Present

Lead year-round technology operations for THON, which raises $10M+ annually: maintain the organization's primary website, ship new features, and oversee the technical systems the organization relies on.

Penn State Advanced Vehicle Team (AVT) Project Manager Fall 2026 – Present

Manage planning and deliverables for Penn State's team in Year 1 of the EcoCAR Challenge (managed by Argonne National Laboratory; sponsored by GM and MathWorks), building a fully electric vehicle while establishing the structure and processes that later teams will build on.

Association for Computing Machinery (ACM) Web & Design Director Apr 2026 – Present

Republic of Korea Air Force Squad Leader Sept 2022 – June 2024

Relevant Coursework

Computer Science

- CMPSC 131: Programming and Computation I
- CMPSC 132: Programming and Computation II
- CMPSC 221: Object Oriented Programming
- CMPSC 311: Introduction to Systems Programming

- CMPSC 360: Discrete Mathematics
- CMPSC 461: Programming Language Concepts
- CMPSC 464: Theory of Computation
- CMPSC 465: Data Structures and Algorithms

Computer Engineering

- CMPEN 270: Digital Design
- CMPEN 331: Computer Organization and Design

Awards and Honors

1st Place, ThinkNeuro LLC Project Competition	Apr 2026
1st Place, IEEE Student Chapter Battle Bots Competition	Apr 2026
PIT-UN Funding Award ($3,750)	Apr 2025
3rd Place, ACM Pathfinder Robotics Challenge	Apr 2025
Republic of Korea Air Force Commendation Award	Aug 2023

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL

Frameworks and Tools: React, Node.js, NestJS, Express, Flask, MongoDB, Redis, Firebase, Kubernetes, Git, AWS Bedrock, DuckDB

Machine Learning: TensorFlow, PyTorch, FAISS, Mem0, SHAP, spaCy, Scikit-learn, Flower